

MB 521: Disruptive Technologies for Data Science

Course Code	MB 521
Course Name	Disruptive Technologies for Data Science
L-T-P-C	2-0-0-2
Prerequisites	None
Intended for	MBA
Mutual Exclusion	None
Approval	52nd BoA (Nov 2023)

Course Contents

- **Overview of Disruptive Technologies:** Waves of Technology evolution; Digitization and digitalization; Digitalization and disruption; Disruptive innovation; The waves of internet and related technological advancements; Networks, 5G and sensors. (5 sessions)
- **Computing Technologies:** Moore's Law and its impact, miniaturization of computers, Quantum computing and its prospects in business. (3 sessions)
- **Artificial Intelligence:** Concepts of Artificial Intelligence (AI), Integrating AI into human world, AI and robotics, virtual and augmented reality, Impact of AI and robotics in business, AI Entrepreneurs, AI Disruption. (4 sessions)
- **Blockchain and Bitcoin:** Blockchain concepts, Blockchain and Coded currency, Blockchain and Enterprise, Blockchain as a technology of trust, Blockchain driving the business and beyond. (3 sessions)
- **Internet of Things and Cloud Computing:** Evolution of Internet of Things, Economic Impact of IoT, IoT as a revolutionary technology, IoT Challenges, risks and dangers; Internet of Things to Internet of Everything, Basics of cloud computing, Cloud computing and web 2.0, Ways to cloud compute and its business benefits, Personal cloud, Edge computing. (4 sessions)
- **Multidimensional Printing:** Additive and subtractive manufacturing, Decentralizing and disrupting manufacturing, Mass customization, Barriers to additive manufacturing. (2 sessions)
- **Nanotechnology:** New materials and their applications, nanotechnology initiatives. (2 sessions)
- **Biotechnology and Neurotechnology:** Importance of biotechnologies and neurotechnologies, Biotechnology applications in the fields of medicine to manufacturing, workings of neurotechnologies and their impact. (3 sessions)
- **Clean Energy Technology and Geo Engineering:** Clean energy, and its distribution and storage technologies, Geo engineering and global warming. (2 sessions)

Text Books

NA

References

1. Klaus Schwab, **Shaping the Future of the Fourth Industrial Revolution**, Penguin Random House, 2018.
2. Daniel Franklin, **Megatech: Technology in 2050**, Profile Books Ltd, (The Economist), 2017.
3. Steve Case, **The Third Wave**, Simon & Schuster Paperbacks, 2016
4. Christopher Barnatt, **A Brief Guide to Cloud Computing**, Constable & Robinson Ltd., 2010.
5. Peter H. Diamandis and Steven Kotler, **The Future is Faster Than You Think: How Converging Technologies are Transforming Business, Industries and Our Lives**, Simon & Schuster Paperbacks, 2020
6. Peter Thiel, **Zero to One: Notes on Startups or How to Build the Future**, Penguin Random House (Virgin Books), 2014.
7. Jean-Marie Dru, **The new Ways to New**, Wiley India Pvt. Ltd, 2015
8. Clayton M. Christensen, **The Innovator's Dilemma**, Collins Business Essentials, 2006
9. Peter H. Diamandis, Steven Kotler, **Bold: How to Go Big, Create Wealth and Impact the World**, Simon & Schuster, 2015.
10. Henry Chesbrough, **Open Innovation: The New Imperative for Creating and Profiting from Technology**, Harvard Business School Press, 2006.
11. Daniel Kellmerrit, Daniel Obodovski, **The Silent Intelligence: The Internet of Things**, DnD Ventures, 2013
12. Alec Ross, **The Industries of the Future**, Simon & Schuster UK Ltd., 2016
13. Jamie Bartlett, **The People vs. Tech: How the Internet is Killing Democracy**, Penguin Random House, 2018